Establishing a human chorionic gonadotropin cutoff to guide methotrexate treatment of ectopic pregnancy: a systematic review

Seema Menon, M.D.,[a] *John Colins, M.D.,*[b] *and Kurt T. Barnhart, M.D., M.S.C.E.*[a,c]

[a] Penn Fertility Care, Department of Obstetrics and Gynecology, and [c] Center for Clinical Epidemiology and Biostatistics, University of Pennsylvania, Philadelphia, Pennsylvania; and [b] Department of Obstetrics and Gynecology, McMaster University, Hamilton, Ontario, Canada

Objective: Successful medical management of an ectopic pregnancy is inversely associated with initial hCG level. The purpose of this publication is to assess whether there is a level of hCG above which failure rate substantially increases.

Design: A systematic review and summary analysis was performed, including studies reporting methotrexate treatment outcomes as stratified by various hCG ranges.

Setting: Academic medical center.

Patient(s): Review of published information regarding patients treated with methotrexate.

Intervention(s): None.

Main Outcome Measure(s): Success and failure rate of medical management.

Result(s): Five observational studies, including 503 women, were found that reported successes in using single-dose methotrexate stratified by initial hCG concentration. Failure rates increase with increasing hCG levels. A substantial and statistically significant increase in failure rates is seen when comparing patients who have initial hCG levels of >5,000 mIU/mL with those who have initial levels of <5,000 mIU/mL (odds ratio: 5.45; 95% confidence interval: 3.04, 9.78). The failure rate for women who had an initial concentration between 5,000 and 9,999 mIU/mL was significantly higher than that for those who had initial levels between 2,000 and 4,999 mIU/mL (odds ratio: 3.76; 95% confidence interval: 1.16, 12.33).

Conclusion(s): Results support a substantial increase in failure of medical management with single-dose methotrexate when the initial hCG is above 5,000 mIU/mL. Methotrexate should be used with caution in patients with ectopic pregnancy who present with hCG levels above this level. (Fertil Steril® 2007;87:481–4. ©2007 by American Society for Reproductive Medicine.)

Key Words: Methotrexate, hCG, ectopic pregnancy, treatment, failure

Methotrexate has been established as an alternative to surgery for the treatment of ectopic pregnancy. Much focus has been placed on improving success rates, although multiple publications have supported its safety and efficacy over the years (1–4). The impact of appropriate patient selection on successful treatment outcomes has been emphasized. This has served as the impetus behind several investigations that have been aimed to identify prognostic indicators of successful therapy. Many variables have been correlated with success rates, including the presence of a yolk sac on ultrasound, size of ectopic mass, and hCG levels (5–7). Although there is consensus regarding the relationship between lower hCG levels and lower treatment failure rates, no agreement has been reached regarding an absolute hCG level that serves as a relative contraindication for methotrexate therapy. Multiple studies have published data that are useful in answering this question, but singly, none have been definitive. An analysis of these data is performed to assess whether there is a level of hCG value above which methotrexate should be used with caution.

Received June 28, 2006; revised October 5, 2006; accepted October 10, 2006.

Reprint requests: Kurt T. Barnhart, M.D., M.S.C.E., Penn Fertility Care, Department of Obstetrics and Gynecology, University of Pennsylvania Medical Center, 3701 Market Street, Suite 810, Philadelphia, PA 19104 (FAX: 215-615-4200; E-mail: kbarnhart@mail.obgyn.upenn.edu).

MATERIALS AND METHODS

An OVID MEDLINE search (1966 to present) was conducted by using the keywords “methotrexate,” “hCG,” and “ectopic pregnancy” to find all pertinent publications. One hundred seventy-eight publications were reviewed for eligibility. Publications that were included in this analysis reported the number of patients successfully and unsuccessfully treated within various hCG ranges. The single-dose methotrexate protocol was followed in all of the

0015-0282/07/$32.00
doi:10.1016/j.fertnstert.2006.10.007

TABLE 1

Summary of publications.

Author	No. of patients with outcomes reported by hCG range	Ranges of hCG used for comparison
Elito et al. (8)	40	<1,500 1,500–5,000 5,000
Lipscomb et al. (9)	353	<1,000 1,000–1,999 2,000–4,999 5,000–9,999 10,000–14,999 ≥15,000
Tawfiq et al. (10)	57	<1,000 1,000–1,999 2,000–3,999 4,000–6,000 6,000–8,000 >8,000
El-Lamie et al. (11)	3	<1,000 1,000–10,000 >10,000
Gamzu et al. (12)	50	<2,000 ≥2,000

Menon. Improving methotrexate treatment success. Fertil Steril 2007.

selected publications. Those reporting treatment outcomes using other treatment protocols such as local (intratubal) injection were not included (7). Ultimately, data from five publications were analyzed (8–12). Although many publications have correlated lower hCG levels with lower treatment failure rates, the majority have presented mean hCG levels of those successfully and unsuccessfully treated; breakdown of success rates within hCG ranges was not reported (6, 13–17).

The publications from which data were extracted reported results of observational studies. Therefore, a complete meta-analysis was not performed. Guidelines for analyzing data from observational studies have been established and were used in this analysis (18). Data were synthesized between publications by summation. First, the total number of patients included in all five publications was calculated. Treatment outcomes were reported by hCG ranges (Table 1). Broader hCG ranges also were created in addition to the ones reported. These broader ranges compared treatment outcomes of patients above and below a defined hCG value; for example, patients with hCG values of ≥1,000 mIU/mL were compared with those with hCG values of <1,000 mIU/mL. A total of 26 hCG ranges were developed.

For every hCG range, the total number of patients and the total number of patients successfully and unsuccessfully treated was calculated. A considerable amount of overlap existed between the publications. In these cases, treatment outcomes from different studies were combined.

Success and failure rates also were calculated for every hCG range, and 95% confidence intervals were determined for failure rates. The risk of failed treatment was compared between subsequent hCG ranges by calculating odds ratios, 95% confidence intervals, and P values. All statistical tests were performed by using STATA software (Stata Park, NC).

RESULTS

Failure rates increased with increasing hCG levels (Table 2 and Fig. 1). To obtain meaningful comparisons with sufficient sample size, data were collapsed into categorical ranges for initial hCG concentration (Table 2). The total number of women included in this analysis was 503. We assessed for differences in failure rates among successive strata as well as above and below a certain cut point. For example, we compared failure rates between patients with hCG levels above and below 1,000 mIU/mL, and we compared those with hCG levels of <1,000 mIU/mL with those whose levels were in the 1,000- to 1,999-mIU/mL range.

TABLE 2

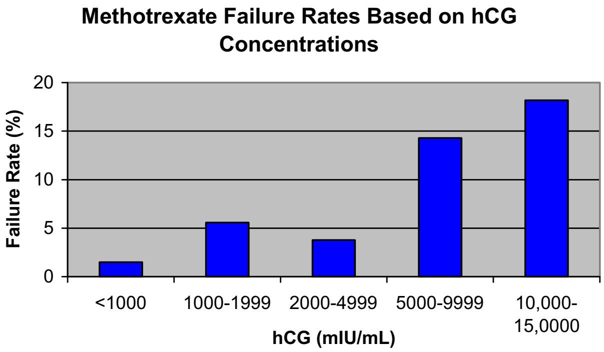
Treatment outcome results of five hCG ranges.

hCG (mIU/mL)	Total no. of patients	Failure (no. of patients)	Success (no. of patients)	Success rate (%)	Failure rate (%)	95% Confidence interval
<1,000	135	2	133	98.52	1.48	0.18–5.24
1,000–1,999	54	3	51	94.44	5.56	1.16–15.39
2,000–4,999	106	4	102	96.23	3.77	1.04–9.38
5,000–9,999	49	7	42	85.71	14.29	5.94–27.24
10,000–150,000	22	4	18	81.82	18.18	5.19–40.28

Menon. Improving methotrexate treatment success. Fertil Steril 2007.

FIGURE 1

Relationship between hCG levels and failure rates.



Menon. Improving methotrexate treatment success. Fertil Steril 2007.

The results showed marked increase in failure rates at levels of >5,000 mIU/mL (Table 2). A sharp increase in failure rates was seen when comparing patients with an initial hCG value in the 2,000- to 4,999-mIU/mL range with those with an initial value in the 5,000- to 9,999-mIU/mL range. Treatment-failure odds-ratio calculations of the five narrow hCG ranges confirmed the significance of poorer outcome among those with hCG values of >5,000 mIU/mL. The odds ratios peaked when comparing the 2,000- to 4,999-mIU/mL hCG range with that of the 5,000- to 9,999-mIU/mL range; this is the only comparison that yielded statistically significant results (Table 3). In other words, a patient is 3.76-fold more likely to fail therapy with an initial hCG level of 5,000–9,999 mIU/mL, compared with an initial hCG level of 2,000–4,999 mIU/mL. For every 10 treatments, there will be one more failure if the hCG level is 5,000–9,999 than there would be if it is 2,000–4,999. (The rate difference in Table 2 is 14.29–3.77, or ~10%.)

TABLE 3

Risk of treatment failure comparison between patients divided into narrow hCG ranges.

hCG level (IU/L)	Odds ratio (95% confidence interval)	*P* value
<1,000 vs. 1,000–1,999	3.75 (0.64, 21.82)	.12
1,000–1,999 vs. 2,000–3,999	2.57 (0.47, 13.93)	.27
2,000–4,999 vs. 5,000–9,999	3.76 (1.16, 12.33)	.02
5,000–9,999 vs. 10,000–14,999	1.27 (0.41, 3.90)	.67

Menon. Improving methotrexate treatment success. Fertil Steril 2007.

TABLE 4

Risk of treatment failure comparison between patients divided into broad hCG ranges.

Level of hCG (mIU/mL)	Odds ratio (95% confidence interval)
<5,000 vs. >5,000	5.45 (3.04, 9.78)
<8,000 vs. >8,000	3.78 (2.19, 6.52)
<10,000 vs. >10,000	3.00 (1.68, 5.36)

Note: For each category, *P*<.01.

Menon. Improving methotrexate treatment success. Fertil Steril 2007.

Comparisons between broader hCG values also support the idea that failure rates significantly increase at levels of >5,000 mIU/mL. The risk of failing treatment decreases when comparing two groups with a cutoff of >5,000 mIU/mL (Table 4). This suggests that little difference is seen when comparing failure rates between groups with a cutoff of >5,000 mIU/mL, because the overall failure rate of both groups is increased.

DISCUSSION

The impact that patient selection has on methotrexate success is accepted widely. Guidelines published by American College of Obstetricians and Gynecologists help direct appropriate patient selection (19). The absolute indications and contraindications largely focus on patient safety. The relative indications and contraindications are mainly concerned with treatment efficacy. Serum hCG is one variable that is considered under these guidelines. Rather than specifying an absolute number as a cutoff, a range between 6,000 and 15,000 mIU/mL is suggested (19). Wider variation is found in the literature, with hCG-cutoff suggestions ranging from 1,500 mIU/mL to 9,000 mIU/mL (15, 16).

Methodological difficulties in this study included the large number of different cutoffs that are used to stratify data in the literature and the large range of hCG values reported. We chose five categories to summarize, because these ranges are clinically meaningful and represented a relatively equal distribution of the data. A second difficulty in interpretation of the findings of these analyses is that results could be skewed by the cut point chosen. For example, when comparing the failure rates above and below 1,000 mIU/mL, the median hCG level in the group with level >1,000 mIU/mL may be anywhere from just over 1,000 mIU/mL to well over 10,000 mIU/mL. This undoubtedly would affect the analysis. Alternatively, direct comparison of two strata may suffer from a small sample size and from insufficient power to detect a statistical significant difference in failure rates between the two groups. We chose to analyze the data summarized in both ways. Results of both analyses suggest that failure rates are substantially and statistically increased when the initial hCG value is >5,000 mIU/mL.

It is important to note that failure of treatment was defined in all studies as requirement of surgical therapy after methotrexate therapy was begun. However, one publication allowed only for one dose of methotrexate (8). If no appropriate drop in hCG level was seen between days 4 and 7, the patient was surgically treated. In the other four studies, multiple doses of methotrexate were allowed before surgery was undertaken. When data analysis was performed excluding the study that used a stricter definition of treatment failure, similar findings resulted. The failure rate increased sharply between 2,000 and 4,999 mIU/mL and between 5,000 and 9,999 mIU/mL: 3.77% (95% confidence interval: 1.04%, 9.38%) vs. 14.29% (95% confidence interval: 5.94%, 27.24%). Odds ratios comparing failed treatments between these two hCG ranges were also the only statistically significant findings noted, after comparisons were made between the narrow hCG ranges: 3.79 (95% confidence interval: 1.16, 12.33; $P = .018$).

Results of this analysis support caution when using the single-dose methotrexate protocol to medically treat a woman for ectopic pregnancy who has an initial hCG of >5,000 mIU/mL. Interestingly, this level has been suggested as a cutoff in a review published elsewhere; however, the investigator did not present the method by which this level was selected (20). Although this analysis included only five studies, the experience of a total of 500 patients treated with single-dose methotrexate was taken into consideration. It is notable that all studies summarized used single-dose methotrexate. It is not known whether results would differ for the use of multidose or two-dose (21) protocols. Multidose protocols have been shown to have a lower failure rate and have been used for women with a higher initial hCG value (22). A clinician should proceed with caution when treating a patient with ectopic pregnancy and an initial hCG level of >5,000 mIU/mL.

REFERENCES

1. Stovall TG, Ling FW, Carson SA, Buster JE. Nonsurgical diagnosis and treatment of tubal pregnancy. Fertil Steril 1990;54:537–8.
2. Stovall TG, Ling FW. Single-dose methotrexate: an expanded clinical trial. Am J Obstet Gynecol 1993;168:1759–65.
3. Lipscomb GH, Derita B, McCord ML, Portera CJ, Ling FL. Analysis of three hundred fifteen ectopic pregnancies treated with single-dose methotrexate. Am J Obstet Gynecol 1998;178:1354–8.
4. Stovall TG, Ling FW, Gray LA. Single-dose methotrexate for treatment of ectopic pregnancy. Obstet Gynecol 1991;77:754–7.
5. Bixby S, Tello R, Kuligowska E. Presence of a yolk sac on transvaginal sonography is the most reliable predictor of single-dose methotrexate treatment failure in ectopic pregnancy. J Ultrasound Med 2005;24:591–8.
6. Potter MB, Lepine LA, Jamieson DJ. Predictors of success with methotrexate treatment of tubal ectopic pregnancy at Grady Memorial Hospital. Am J Obstet Gynecol 2003;188:1192–4.
7. Nazac A, Gervaise A, Bouyer J, De Tayrac R, Capella-Allouc S, Fernandez H. Predictors of success in methotrexate treatment of women with unruptured tubal pregnancies. Ultrasound Obstet Gynecol 2003;21:181–5.
8. Elito J Jr, Reichmann AP, Uchiyama MN, Camano L. Predictive score for the systemic treatment of unruptured ectopic pregnancy with a single dose of methotrexate. Int J Gynaecol Obstet 1999;67:75–9.
9. Lipscomb GH, McCord ML, Stovall TG, Huff G, Portera SG, Ling FW. Predictors of success of methotrexate treatment in women with tubal ectopic pregnancy. N Engl J Med 1999;341:1974–8.
10. Tawfiq A, Agmeya AF, Claman P. Predictors of treatment failure for ectopic pregnancy treated with single-dose methotreaxte. Fertil Steril 2000;74:877–80.
11. El-Lamie I, Shehata NA, Kamel HA. Intramuscular methotrexate for tubal pregnancy. J Reprod Med 2002;47:144–50.
12. Gamzu R, Almog B, Levin Y, Avni A, Jaffa A, Lessing JB, et al. Efficacy of methotrexate treatment in extrauterine pregnancies defines by stable or increasing human chorionic gonadotropin concentrations. Fertil Steril 2002;77:761–5.
13. Stika CS, Anderson L, Frederiksen MC. Single-dose methotrexate for the treatment of ectopic pregnancy: Northwestern Memorial Hospital three-year experience. Am J Obstet Gynecol 1996;174:1840–8.
14. Cassik P, Ofili-Yebovi D, Yazbek J, Lee C, Elson J, Jurkovic D. Factors influencing the success of conservative treatment of interstitial pregnancy. Ultrasound Obstet Gynecol 2005;26:279–82.
15. Dudley PS, Heard MJ, Sangi-Haghpeykar H, Carson SA, Buster JE. Characterizing ectopic pregnancies that rupture despite treatment with methotrexate. Fertil Steril 2004;82:1374–8.
16. Corsan GH, Karacan M, Qasim S, Bohrer MK, Ranson MX, Kemmann E. Identification of hormonal parameters for successful systemic single-dose methotrexate therapy in ectopic pregnancy. Hum Reprod 1995;10:2719–22.
17. Lecuru F, Robin F, Bernard JP, Maizan de Malartic C, Mac-Cordick C, Boucaya V, et al. Single-dose methotrexate for unruptured ectopic pregnancy. Int J Gynaecol Obstet 1998;61:254–9.
18. Stroup DF, Berlin JA, Morton SC, Olkin I, Williamson GD, Rennie D, et al. Meta-analysis of observational studies in epidemiology a proposal for reporting. JAMA 2000;283:2008–12.
19. American College of Obstetricians and Gynecologists. ACOG practice bulletin. Medical management of tubal pregnancy. Number 3, December 1998. Clinical management guidelines for obstetrician–gynecologists. Int J Gynaecol Obstet 1999;65:97–103.
20. Farquhar CM. Ectopic pregnancy. Lancet 2005;366:583–91.
21. Barnhart K, Hummel AC, Sammel MD, Menon S, Jain J, Chakhtoura N. Use of “2-dose” regimen of methotrexate to treat ectopic pregnancy. Fertil Steril. In press.
22. Barnhart KT, Gosman G, Asby R, Sammel MD. The medical management of ectopic pregnancy: a meta-analysis comparing “single dose” and “multidose.” Obstet Gynecol 2003;101:778–84.